

Logistic Regression

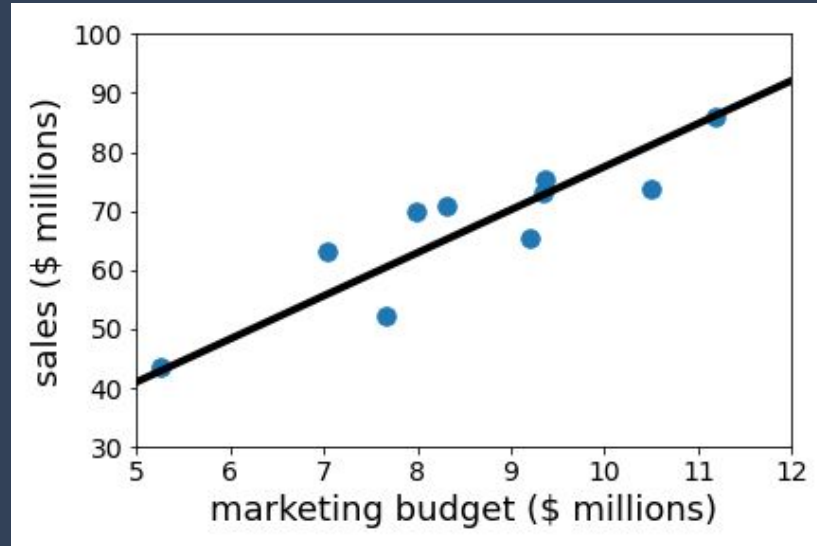


**INTERVIEW
KICKSTART**



**INTERVIEW
KICKSTART**

Regression vs. Classification



Regression vs. Classification

Typical Regression problems:

- Predict house sales price (Zillow, Freddie Mac, ...)

Regression vs. Classification

Typical Regression problems:

- Predict house sales price (Zillow, Freddie Mac, ...)
- Predict stock market price (Quant trading)

Regression vs. Classification

Typical Regression problems:

- Predict house sales price (Zillow, Freddie Mac, ...)
- Predict stock market price (Quant trading)
- Forecast macroeconomic indicators (Gov, Banks, ...)

Regression vs. Classification

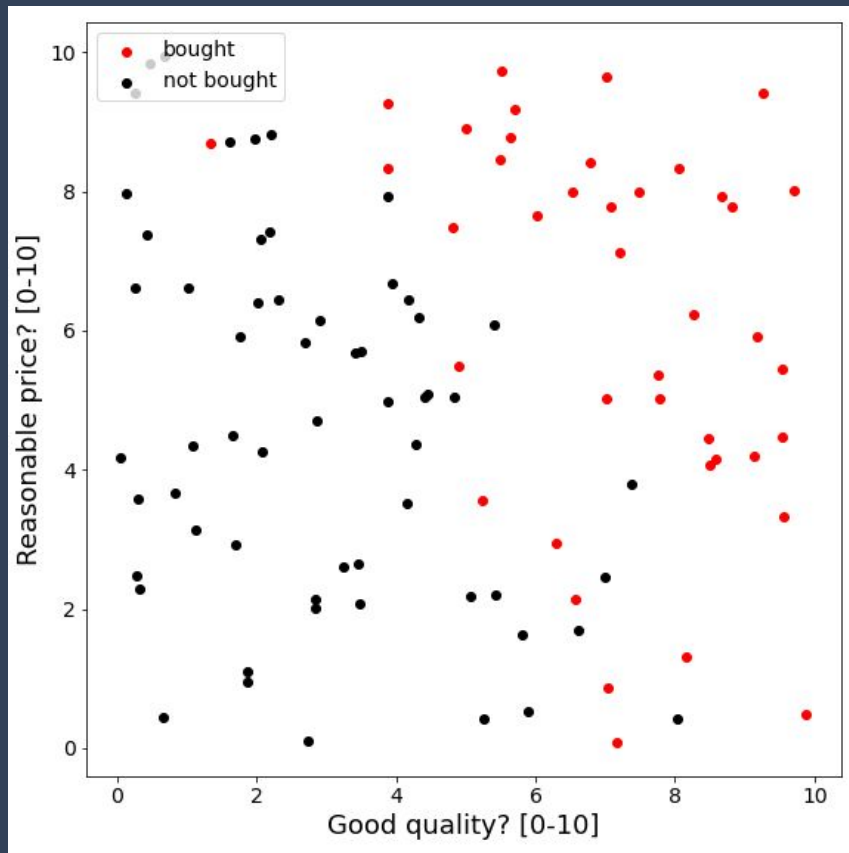
Typical Regression problems:

- Predict house sales price (Zillow, Freddie Mac, ...)
- Predict stock market price (Quant trading)
- Forecast macroeconomic indicators (Gov, Banks, ...)
- Forecast data volume growth, user growth (Tech company data centers)

Problem Statement

Good quality? [0-10]	Good price? [0-10]	Bought the product
3.2	4.4	0
1.2	3.1	0
4.1	8.5	1
...	...	...
4.2	3.6	0

Problem Statement



Regression vs. Classification

Typical Classification problems:

- Predict if an email is spam (Google, ...)

Regression vs. Classification

Typical Classification problems:

- Predict if an email is spam (Google, ...)
- Predict if a customer will buy certain product (Amazon, ...)

Regression vs. Classification

Typical Classification problems:

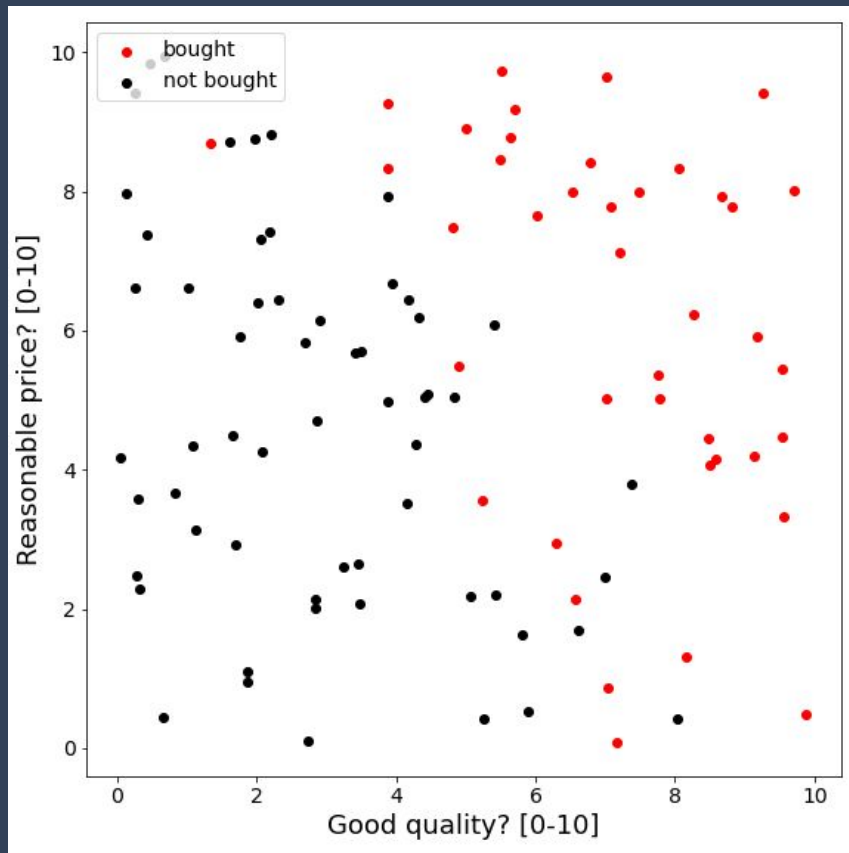
- Predict if an email is spam (Google, ...)
- Predict if a customer will buy certain product (Amazon, ...)
- Predict if a transaction is fraudulent (credit card companies, Amazon, ...)

Regression vs. Classification

Typical Classification problems:

- Predict if an email is spam (Google, ...)
- Predict if a customer will buy certain product (Amazon, ...)
- Predict if a transaction is fraudulent (credit card companies, Amazon, ...)
- Predict cancer/no cancer from medical imaging scans (Healthcare)

Problem Statement



Logistic Regression

- Binary classification problem

Logistic Regression

- Binary classification problem
 $y = 0$ (not buy/negative class), or
 $y = 1$ (buy/positive class)

Logistic Regression

- Binary classification problem
 $y = 0$ (not buy/negative class), or
 $y = 1$ (buy/positive class)
- What if we apply our linear regression?
Fit to a linear equation, $y = Xw$?

Logistic Regression

- Binary classification problem
 $y = 0$ (not buy/negative class), or
 $y = 1$ (buy/positive class)
- What if we apply our linear regression?
Fit to a linear equation, $y = Xw$?
 $-\infty < y < +\infty$

Logistic Regression

- Binary classification problem
 $y = 0$ (not buy/negative class), or
 $y = 1$ (buy/positive class)
- What if we apply our linear regression?
Fit to a linear equation, $y = Xw$?
 $-\infty < y < +\infty$
- What would be an ideal prediction?

Logistic Regression

- Binary classification problem
 $y = 0$ (not buy/negative class), or
 $y = 1$ (buy/positive class)
- What if we apply our linear regression?
Fit to a linear equation, $y = Xw$?
 $-\infty < y < +\infty$
- What would be an ideal prediction?
 $0 \leq y \leq 1$

The logistic function

Two steps:

- $z = Xw$, linear regression

The logistic function

Two steps:

- $z = Xw$, linear regression
- A function that turns $z (-\infty, +\infty)$ to $P(y=1 \mid X, w) (0, 1)$

The logistic function

Two steps:

- $z = Xw$, linear regression
- A function that turns z $(-\infty, +\infty)$ to $P(y=1 \mid X, w)$ $(0, 1)$

$$S(x) = \frac{1}{1 + e^{-x}}$$

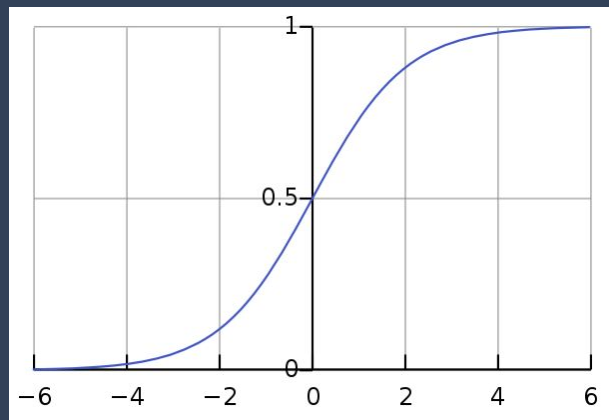
The logistic function

Two steps:

- $z = Xw$, linear regression
- A function that turns $z (-\infty, +\infty)$ to $P(y=1 \mid X, w) (0, 1)$

$$S(x) = \frac{1}{1 + e^{-x}}$$

Logistic function



The logistic function

Two steps:

- $z = Xw$, linear regression
- A function that turns z $(-\infty, +\infty)$ to $P(y=1 \mid X, w)$ $(0, 1)$

$$P(y = \text{Will buy} \mid \text{person } X) = \frac{1}{1 + e^{-(X \cdot w)}}$$
$$P(y = \text{Won't buy} \mid \text{person } X) = 1 - \frac{1}{1 + e^{-(X \cdot w)}}$$

The logistic function

Two steps:

- $z = Xw$, linear regression
- A function that turns z $(-\infty, +\infty)$ to $P(y=1 \mid X, w)$ $(0, 1)$

$$P(y = \text{Will buy} \mid \text{person } X) = \frac{1}{1 + e^{-(X \cdot w)}}$$
$$P(y = \text{Won't buy} \mid \text{person } X) = 1 - \frac{1}{1 + e^{-(X \cdot w)}}$$

$P(y=1 \mid X, w)$: probability of $y=1$ given X , parametrized by w

The logistic function

Two steps:

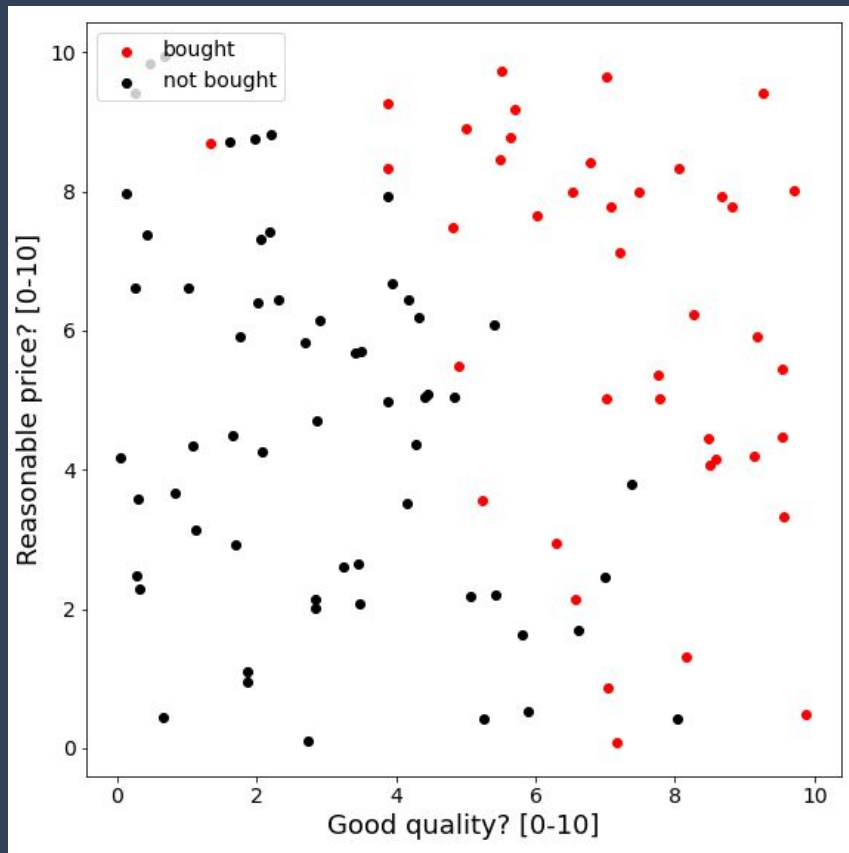
- $z = Xw$, linear regression
- A function that turns z $(-\infty, +\infty)$ to $P(y=1 \mid X, w)$ $(0, 1)$

$$P(y = \text{Will buy} \mid \text{person } X) = \frac{1}{1 + e^{-(X \cdot w)}}$$
$$P(y = \text{Won't buy} \mid \text{person } X) = 1 - \frac{1}{1 + e^{-(X \cdot w)}}$$

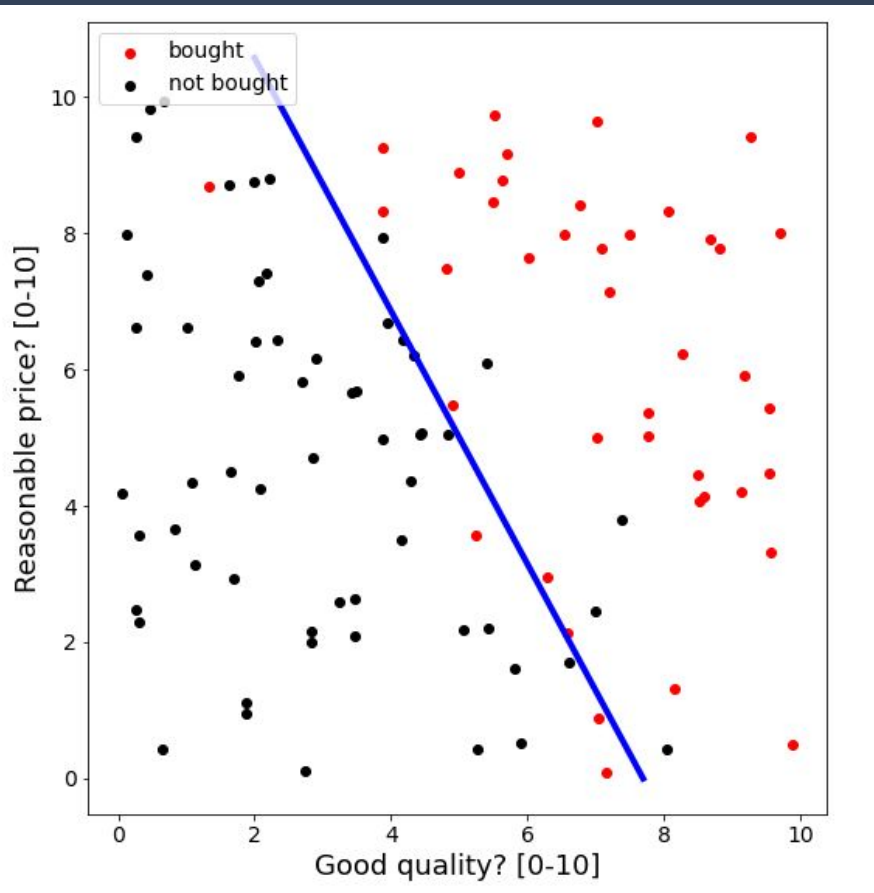
$P(y=1 \mid X, w)$: probability of $y=1$ given X , parametrized by w

$$P(y=1 \mid X, w) + P(y = 0 \mid X, w) = 1$$

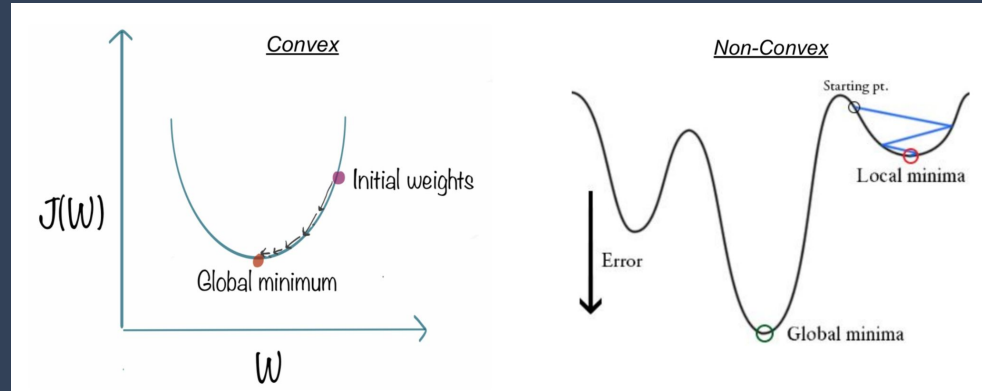
Decision Boundary



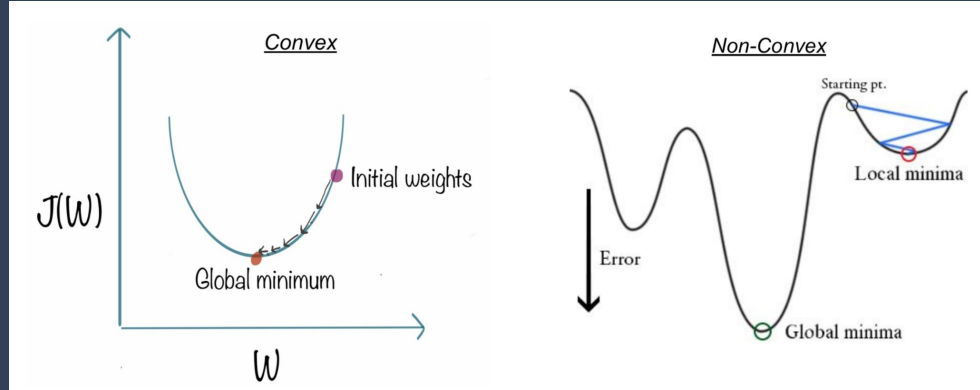
Decision Boundary



Loss function

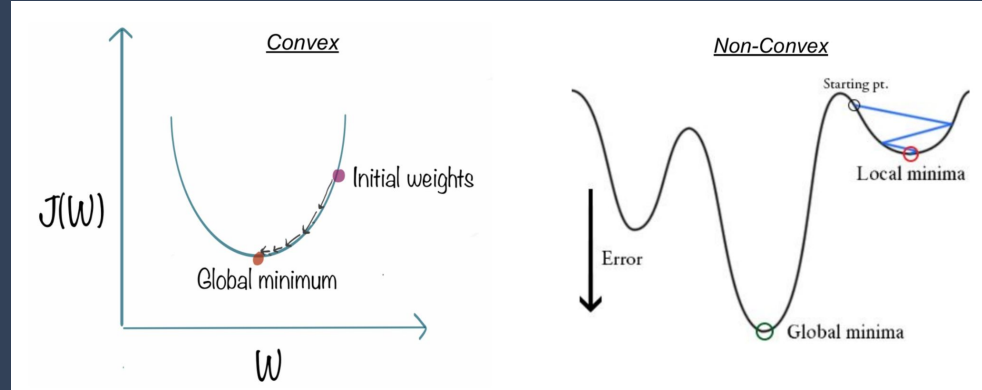


Loss function



- Log loss/binary cross-entropy loss:
 - MSE becomes non-convex after applying the sigmoid function

Loss function

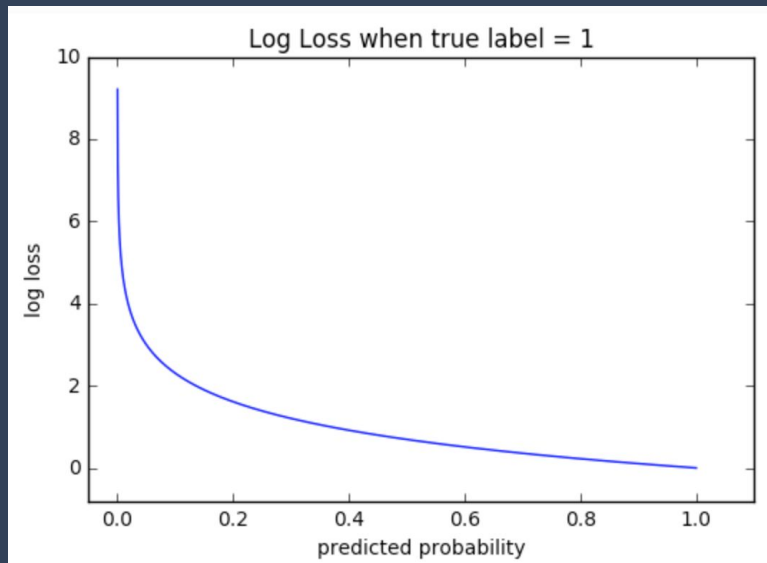


- Log loss/binary cross-entropy loss:
 - MSE becomes non-convex after applying the sigmoid function
 - Log loss:

$$LogLoss = -\frac{1}{n} \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

The log loss function

$$\text{LogLoss} = -\frac{1}{n} \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$



The log loss function

$$LogLoss = -\frac{1}{n} \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

- Log loss penalizes on confident + wrong predictions, which leads to well behaved probabilistic outputs, sometimes at the cost of accuracy.

The log loss function

$$LogLoss = -\frac{1}{n} \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

- Log loss penalizes on confident + wrong predictions, which leads to well behaved probabilistic outputs, sometimes at the cost of accuracy.
- Loss loss is also called the binary cross-entropy, which is a special case for cross-entropy.

The log loss function

$$\text{LogLoss} = -\frac{1}{n} \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

- Log loss penalizes on confident + wrong predictions, which leads to well behaved probabilistic outputs, sometimes at the cost of accuracy.
- Loss loss is also called the binary cross-entropy, which is a special case for cross-entropy.
 - Entropy (p): measures uncertainty (information)

The log loss function

$$\text{LogLoss} = -\frac{1}{n} \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

- Log loss penalizes on confident + wrong predictions, which leads to well behaved probabilistic outputs, sometimes at the cost of accuracy.
- Loss loss is also called the binary cross-entropy, which is a special case for cross-entropy.
 - Entropy (p): measures uncertainty (information)
 - Cross-entropy (p, q): information required to convert from p to q.

Multiclass classification



Multiclass classification



Buy Class 1

Not buy Class 0

Multiclass classification

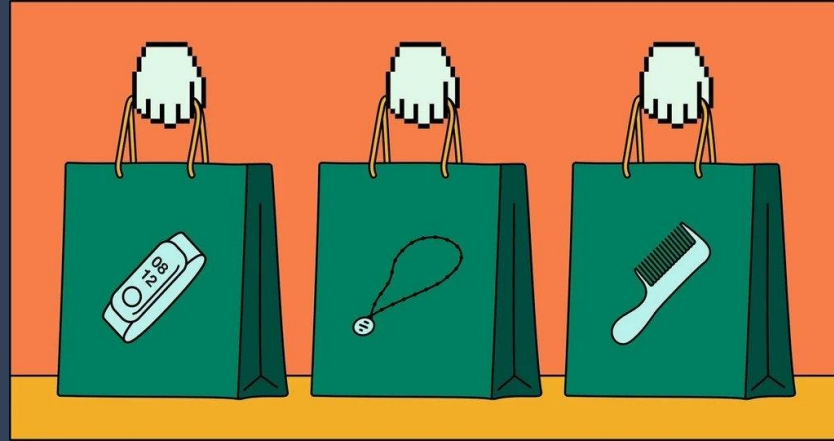


Multiclass classification

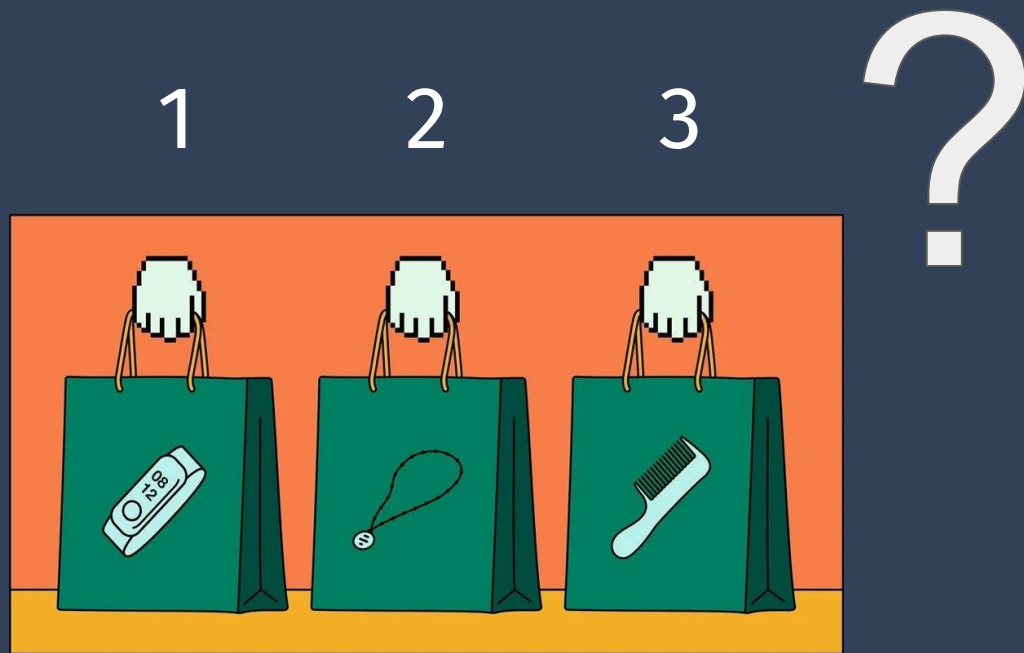
1

2

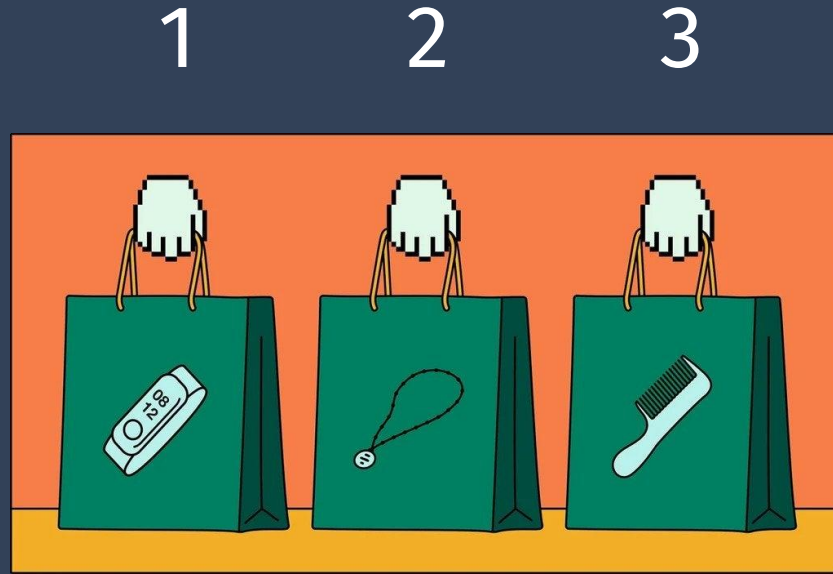
3



Multiclass classification

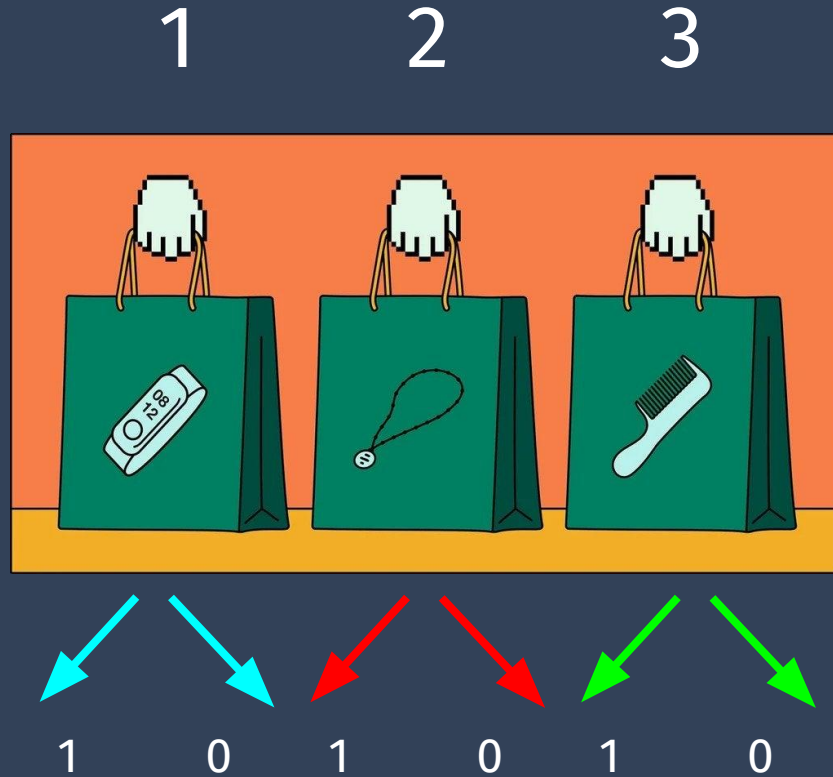


One vs. Rest

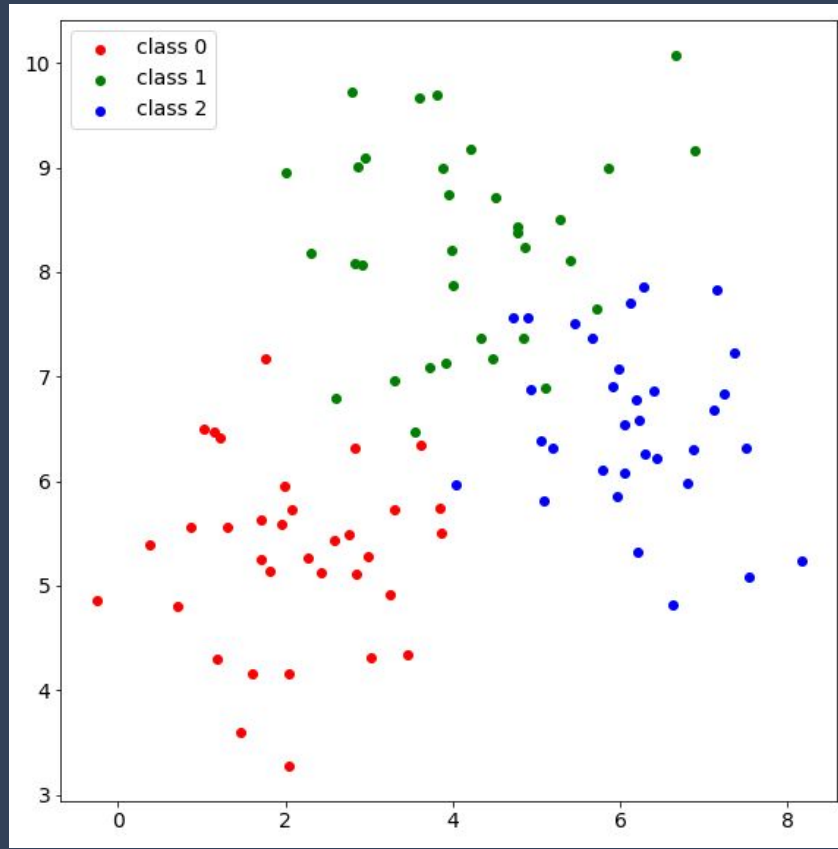


Buy=1 No buy=0

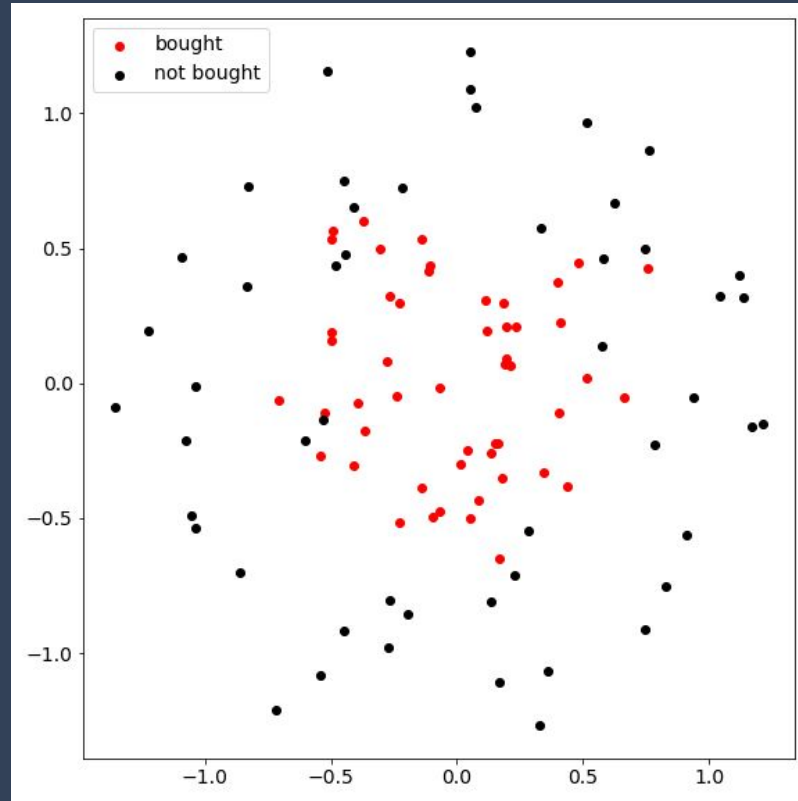
One vs. Rest



One vs. Rest



Polynomial + Regularization



Polynomial + Regularization

- Most of the concept of regularization for linear regression also apply here
- Lasso, Ridge, ElasticNet
- Regularization can be applied to reduce overfitting when we have too complex models and/or too many features